Representations, consciousness and memory in goal-directed behavior

Abstract

Voluntary actions have been characterized as an internal, spontaneous and goal generated behavior. Commonly acknowledged, voluntary actions are embedded within a sequence chain of complex goal-directed actions in a task. Conventional experimental paradigms to study volition, have involved mainly actions that are ‘meaningless’ in nature. To address this limitation of previous studies, in this experiment we proceed with using a problem-solving task: Tower of London (ToL). The subjects move the balls on the peg from the start to the goal configuration. ToL problems were selected to contain a bifurcation point where the participant could choose between 2 pathways by which to solve the problem. Both alternatives pathways had equal value, so there was no particular reason why the participant should choose one option over the other. However, the subject still breaks this symmetry and commits to one of the choices. We believe that the ability to break this symmetry and commit to one choice is what we consider a prototype of a ‘self-initiated’ action. We performed 2 experiments to evaluate whether our choices are dependent on the mental representation of alternatives prior to actions and if the presence of choices could impact the established marker Readiness Potential (RP) (RP is the established biomarker described by the ramping up of negativity which occurs during preparation of a voluntary action). In the first experiment, we included a memory test after participants completed the ToL task to evaluate whether subjects will have a memory representation for alternative ‘unchosen’ configurations when compared to new configurations. The purpose of this memory task was to answer whether the individual represented the other unchosen alternative path B when they committed to path A. If they did represent the ‘unchosen path B’, that means that participants had conscious access to both alternatives when making a decision. And only with the premise that when individuals represent both alternative can we fairly state that they had freedom of choice and were thereby directly responsible for their action consequences. This experiments results would probe the cognitive processes relevant for responsibility as it would add on to the argument that people are morally responsible to their actions as they are indeed representing all alternative when making a decision. In the second experiment, we recorded the Readiness Potential (RP) using EEG, and looked at how it varies depending on the alteration of different independent variables, such as the presence of a TP or not. Our results show mental representation of alternative actions, but we did not find evidence confirming RP being sensitive to this active decision making between the choices.

2. Introduction

2. 1. What is Volition?

The conscious experience of ‘free will’ relies on the ability of an individual to generate actions. It embodies the power of individuals to make decisions, evaluating and choosing between different possible actions and not result from a reflex like response to a stimulus. It is commonly acknowledged by society that only ‘free willed’ or self-initiated acts are conceived as being entitled to the credit or blame. Thereby whether our actions are controlled by ‘ourselves’ is a vital question we must ask to comprehend many human behaviors and reconsider any moral responsibilities (Khalighinejad *et al*., 2018). For instance, for an act to deserve criminal reproach, it must involve both the preceding intention and the physical action. Voluntary action is a crucial component when completing tasks in our daily routine. It gives us a feeling of agency and control over our actions and their consequences, and by that a sense of responsibility of the individual with regards to our society. (Vierkant *et al*, 2019)

Common methodologies to study volition have been constrained in examining internally generate behavior, but more and more evidence from recent studies are suggesting that volition is not merely a single concept, but a set of features (Fried & Haggard, 2017). One of the crucial features of volition is that of goal directedness. For an action to be 'self-generated' it must have a purpose. That means that there is a goal that the movement wants to achieve after execution. Traditional experiments to study volition have already frequently used problems that contain either instructed or free actions. Examples of this include, instructed or non-instructed button presses (Khalighinejad *et al*, 2018). These experiments were widely used in exploring the EEG variability between self-initiated actions and guided instructions in the past few years. However, researchers recently discovered that there is a key limitation to this experimental paradigm (Haggard, 2018). And that is that most of the actions in past experiments were in nature 'meaningless'. That is because the participant lacks a clear purpose when performing the action, therefore the assumption that the actions were 'self-initiated' was widely deceptive itself.

To examine this issue, in this study we used ToL problem-solving tasks. During the active decision process, participants try to match the start configuration of the three colored balls to the goal pattern by moving the balls between the three pegs. There are two main reasons why this experimental paradigm would be a more novel and suitable way to study volition. Firstly, in classical problem solving tasks like ToL, actions involved need to be internally generated as we all self-paced. Even though there may be external stimuli, overall, the participant still needs to work out a way to the solution by internally generating new pieces of information. Furthermore, ToL problems make the participants have a clear objective in mind, this makes each action meaningful as they are all working towards the achievement of the goal. This is something that the classic volition studies lack, but ToL problems addresses: 'actions in ToL are clearly goal directed and included within means end sequences'. And last but most importantly, because we are able to select ToL problems that have many solutions to achieve the same goal, we are able to observe the representation of the alternative means and the active decision between them, which makes this very useful for looking at the feature of motor 'equifinality' in volition. During the problem solving process, in order to ensure that the action is not spontaneously generated and has a planning process prior to execution. Participants of the study were reminded to commit to active planning, as solving the problem in the minimum number of moves may involve monetary compensation. The creation of a standardized task with an endpoint is an example of task setting in which subjects generate their own solution to the tasks. (Shallice *et al*, 1997).

2. 2. 'Self-generated action' in Problem Solving Tasks

In daily life, our experience of volition comes each time we have to face alternative choices, however this can be difficult to translate into an experimental setting (Frith & Haggard, 2018). One of the main issues voluntary action studies have encountered in the past is the lack of a 'purpose of acting'. Many of the experimental paradigms have focused on studying the neural activities preceding the onset of meaningless actions. For instance, in the Kornhuber and Deecke's experiment (Kornhuber & Deecke, 1990), participants were asked to perform free, spontaneous hand movements, during which a novel and peculiar ramping up of negativity was observed preceding each spontaneous action. In the Libet's experiment, apart from studying movement onset, he also recorded the timing of participants 'self-reported urge to move' (Libet *et al*, 1983). The result of this experiment showed that the RP was measured significantly earlier than the intention to move and the timing of movement onset. Therefore, this was interpreted as the subject having unconsciously initiated movement before their conscious will. In both of these experiments, although individuals were given complete freedom to act whenever and as they like, there is a lacking purpose in the nature of the action, which shows poor resemblance to actions humans perform in daily life. RP measured is also highly ambiguous as the recorded neural activity could also just be a result of paying attention or other cognitive functions. Studying volition under the scientific setting that actions are most often 'meaningless' means movements are only voluntary to a small degree. As stated by Wittgeinsten: "voluntary action only becomes meaningful when it is performed in a meaningful context, involving motivation and goal" (Rowe *et al*, 2001).

The definition of volition is not limited to a single idea, instead it is more a coalesce of a set of features, one of the features that we want to explore in this study is that of spontaneity of the action. In a conventional scenario, when the participants are presented with many ways to do a thing, there is no preference for the individual to choose one alternative over the other. However, the subject is still able to break this symmetry and commit to one choice in the process. This is also called Buridan's choice, derived from a paradoxical thought experiment of Buridan's ass. The hungry and thirsty donkey who was given both hay and water, but still dies of hunger and thirst because it couldn't make a rational choice. We suggest that the movement presented when committing to a choice may be a prototype of a self-initiated movement. In order to examine this scenario in an experimental paradigm, we proceeded to use Tower of London problems designed with a Turning Point (TP). In conscious decision making, we expect that the individuals represent both choices before committing to one of them. In this study, we want to examine SOMETHING using a memory test to see whether the subjects actually represent both alternatives when generating a decision. Theoretically, the embedded alternative and the new configurations are both configurations that the participant has never seen. So, we should see no difference between the individual's sensitivity to these two configurations in the memory task. The behavioral study is linked to the idea that we may or may not have the access to that 'process' of achieving the goal. (Fourneret P. & Jeannerod M, 1998)

If we were not to have access to the means of problem-solving, or in other words 'humans are not conscious of the mean, only of our goals' (James, et al 1890), then, the 'alternative path' should not be represented in our memory. However, if the human cognition is sensitive to the problem-solving process, the fact that an alternative path is available should trigger the brain to represent the alternative path even if it ended up unchosen. In other words, even when the participant commits to one path, the alternative remains available to their consciousness.

In the ToL problems, a bifurcation point or Turning Point (TP) were selected. Participants, in theory, have no reason to choose one alternative over the other as both pathways involve minimum number of moves. But still, the participant will commit to one choice to solve the problem. Then, immediately after the subject solves the ToL problem, we show them 7 different configuration of pegs and ask them whether they had encountered them during problem solving. One was the goal, which was used mainly as a control. If the participants are actively engaged with the task, they should always remember the goal configuration. There were 2 old configurations, which were intermediate steps in problem and finally, there were 2 new and 2 alternative configurations. At this point, we ask the participants: "Have you seen this configuration before?". If human cognition did pay attention to the problem-solving process, we expect that there will be some sort of bias in the responses. Participants would more likely respond incorrectly 'seen' to 'alternative' configurations as they may have represented that choice when solving the task.

2.3 Clinical Relevance of Voluntary action studies

The prefrontal cortex (PFC) plays an essential role in organizing goal directed thoughts and behaviors. Connecting reciprocally with most of the cortical and subcortical regions in the brain, the PFC controls many of the key cognitive functions, such as action inhibition, working memory, action monitoring, temporal coding, etc. Patients with lesions in the PFC often have impaired task setting and problem-solving abilities, for instance if the lesion happens in the left PFC, the patient will perform their actions by trial-and-error rather than planning, leading them to take more moves in problem solving tasks than healthy patients (Morris *et al*, 1997). They remain capable of carrying out each action individually, but when it becomes part of a sequence, they struggle or are unable to coordinate the actions in a temporally ordered manner. ToL problems have been used widely in the past to assess the ability of patients to proceed with ordered actions, since ToL portrays a clear resemblance to action planning and monitoring in daily tasks (Szczepanski & Knight, 2014).

The mid-dorsolateral prefrontal cortex (dIPFC) was the region most predominantly linked to action planning function. Past studies using fMRI to image patients during performance of Tower of London tasks has already revealed how lesions to the dIPFC can significantly impair the individuals ability to plan out a sequence of actions and therefore perform coherent execution of complex behaviours. Planning, within the framework of the cascade model, relies largely on episodic control. Each step in the problem demands the participant to constantly update their mental plan to the current restriction encountered. This dimensional integration and consistent evaluation which is crucial in action ability is what makes ToL problems so suitable for studying this function in clinical settings. In each move of ToL, the individual has to update the active task planning paradigm to foresee possible consequences of their current move. (Kaller *et al*, 2011)

Another crucial brain region to voluntary action is the medial frontal cortex. Instead of being involved with planning a sequence of actions, the medial frontal cortex is essential for the initiation of an action. However, on itself the medial frontal cortex is not enough for creation of a movement as studies had shown that its role is mainly in motivating, but not execution or regulation of the initiated action. (Scangos & Stuphorn, 2010). Instead, the supplementary motor are (SMA) and pre-SMA are considered

to have a predominant role in movement initiation and inhibition. Lesions or temporary inactivation using TMS occur to the SEF or SMA regions, the patients present significant impairment of unconscious *involuntary* motor control. Lesions specific to this brain region are rare, but an example to SEF damage was patient JR who presented clear deficits to control voluntary eye movement. He lost the power to switch between saccade plans, as well as prosaccade and antisaccade eye movement toward or away from the stimulus. (Sumner & Husain, 2003)
ToL problems relies on both the cognitive pre-planning of a complex sequence of moves and initiation of the action. Therefore, the ToL experimental paradigm is an innovative and intelligent way that allows us to take a closer look at the nature of voluntary action and the cognitive functions that precede 'self-initiated' moves.

2. 5. Biomarker of Voluntary action: Readiness potential (RP) and its different interpretations

Initially observed by Kornhuber & Deecke, by averaging across several EEG recording epochs to eliminate noise, they observed a gradual accumulation of negativity signals preceding voluntary actions. (Deecke *et al*, 1969). They named this signal: Readiness potential (RP). It was hypothesized that this ramping up of negativity is a build-up in preparation of movement and could be utilized as a biomarker to predict volitional movements (Libet *el al*, 1983). However, the interpretation is varied, some debate that RP accounts for the timing of endogenous generation of a voluntary action, presented greater when in lack of external constraint, while others account it to the planning preceding an action onset. (Parker *et al*, 2020). A recent study looking at the relationship between learning and RP amplitude showed that as participants learned through trials via reinforcement learning, the RP amplitude would gradually increase, proving that RP is associated more closely with the process of 'planning an action', rather than endogenous generation. (Travers *et al*, 2021).

In 1983, Libet and his colleagues carried out a set of experiments to study voluntary action focusing on measuring RP. They recorded the RP ramping to start ~550 msec prior to action onset. But what was astonishing was that the RP preceded the initial intention to move (W) by almost 300ms and the awareness of actually moving (M) by ~420 msec. These results indicate that not only is the voluntary act initiated without conscious awareness of the subject, but the decision to act was later instilled into the person's beliefs, convincing the person that the action was their intentional act. For some time, the argument of Libet's experiment was solid in stating that there is no free will as voluntary actions are in fact initiated before consciousness takes place. (Hari, 2018)

This interpretation has been challenged recently, Schurger's stochastic model rejects RP as a hallmark for voluntary action, rather it argues that RP is just a compilation of fluctuating neural activities that bring the motor activation closer or further from the activation threshold. When these fluctuations end up hitting the threshold, a voluntary action will be triggered. When the activity of many trials is time locked to action and averaged across trials it produces a stochastic model (Schurger A, 2018). The model suggests thereby that there is no neural event triggering RP, instead the onset of RP is just a result of fluctuations of neural activity in the SMA and pre-SMA hitting a threshold. (Schurger *et al*, 2012). From this perspective, we consider the following question: Are self-initiated actions elicited from our own conscious will or are they just random stochastic signals in the brain?

Another argument suggests that RP instead of being a biomarker for voluntary action, it reflects rather the conflict and uncertainty when there is a lack of external triggers affecting 'internally-generated' actions (Nachev et al, 2005). In this cases, the objective is not defined and therefore the RP measured in the medial frontal cortex reflects the uncertainty of when to act, whereas the anterior cingulate reflects the indecision about which action to perform. (Botvinick et al, 2004). If this was true, we would expect a greater RP amplitude for actions that are not defined or 'freer actions'. This is reflected between Intention and guided stimuli trials.

One of the main purposes of this study was to dive into the nature of the RP and provide evidence that RP is not an accumulation of stochastic noises, but rather a neural correlate of voluntary action preceding or even triggering specific, goal directed moves. We hypothesize that there will be a linear correlation between the magnitude of the RP and how free an action is, that is the RP of the actions measured prior to the TP would be greater than that before no TP moves.

The objective of this study is to examine whether the hallmark of internally generated actions: RP will be observed in problem solving tasks like ToL. If seen, we aim to ascertain the interaction of the brain network connectivity with RP for problem solving tasks. And finally, we want to look at how modifying different parameters of ToL may have an impact on the onset timing and amplitude of the RP. For this reason, we selected ToL problem execution conditions involving Intention vs Stimulus. Intention moves are those that are'self-initiated', which require subject pre-planning and active thinking to solve. Whereas the stimulus, on the other hand, comprise of instructed moves where participants need to follow a guided instructed pathway to complete the problem. We hypothesize, that as Intention moves resemble closer a 'self-generated' act, it will have an earlier RP onset than the Stimulus driven actions. This is because, voluntary actions need to be generated entirely from an individual own will with the lack of any external triggers. Then, we also selected problem types either containing alternative paths or no alternative paths. We hypothesize that problems with alternatives will give subjects more freedom of 'choice' , eliciting an 'internally generated action' to take place and thereby resulting in a greater RP amplitude. Other minor factors that were included was for instance, the sequence length of the trial. In which the ToL problems were classified as being either possible be completed in either 3 or 4 moves. This reflects on the planning capacity of the individual as more moves in sequence will require more forward planning.
Another factors of interest, was to study RP amplitude preceding actions within one TP problem. If the TP move was the first step of the problem, the RP amplitude preceding the TP should be greater than any other moves in the problem, because that is supposedly the move preceding decision making that involves clear internal generation from the individual.

I can see you put a lot of more effort in making the HY clearer, well done. I guess you can make them even clearer saying very clearly what we are investigationg and expecting in EXP 1 and separately for EXP 2.

3. Methods

EXPERIMENT 1

3.1. Participant recruitment for behavioral task

25 volunteers were recruited using the UCL psychology subject data pool. Requisites of sign up were age between 18-40 (3 males,22 females, mean age=22.88), right-handed, normal or corrected to normal vision, having no history of psychiatric/psychological disorders or neurological disorders/injuries, including seizures, or any family history of seizures/epilepsy. Participants have to confirm that they are not taking any psychoactive medication, have no history of drug abuse and used any recreational drugs or alcohol 12 hours before the testing session. Participants are compensated for their time with an hourly rate of 9.5 pounds which is approved by the institution. Also, the experimental design and procedure followed the UCL ICN ethics committee (ICN-PH-PWB-20-02-2014c).

3.2. Behavioral task and Procedure

The experiment was programmed in Matlab R2014b using the Psychophysics Toolbox-3 extensions (Brainard, 1997; Kleiner et al., 2007; Pelli, 1997).
After signing the consent form and being explained the experimental procedure, participants in a testing room with a computer and asked to play with the task. The task consisted of a modified version of the of the “Tower of London” (Shallice, 1983). Participants were instructed to move three colored pegs aiming to match the start configuration to the goal. In this experiment, ToL problems were designed to include a TP. The participants were explicitly instructed to plan out their moves as completing the problem in the minimum number of moves possible generated bonus reward. Each of the moves were entailing 2 actions. The actions were in nature self-paced. Before each action, they were presented with a Fixation of cross of 500ms, followed by 2-6s of Jitter. In every move, each finger was corresponding to a ball, and to move the balls, the participant needs to press the corresponding key to the ball they want to move and release the ball by pressing the corresponding key to the location of release.

The ToL problems were selected to contain different conditions to match the cognitive functions examined. Different length problems, either 3 or 4 moves tested for the planning capacity of subjects, while problems that included a TP or no TP examined the ability of participants to represent the ‘unchosen pathway’ when deciding between equally valued alternatives.

Immediately after solving the problem, they had to do memory test. We presented the participants with 7 different configurations and asked: “Have you seen this configuration before?” Participants had to choose either ‘Old’ or ‘New’ depending on whether they thought they had seen the configuration before. The 7 configurations were presented in randomized order and included one control, that is the goal configuration, 2 alternative, 2 old, and 2 new state configurations.
Prior to starting, participants completed one training block to get familiar with the task and check their understanding.

Participants completed a total of 6 blocks. The blocks were presented in a randomized order, with each block consisting of 38 trials, resulting in a total of 228 trials for each participant.

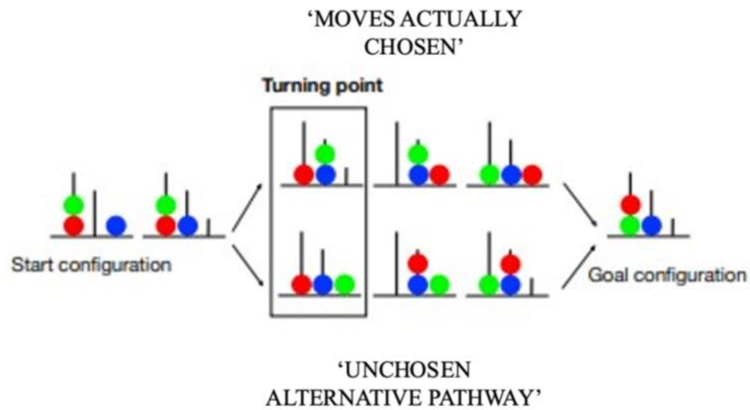

Figure 1: Tower of London (ToL) Problem example with "Turning Point" bifurcation point where the participant needs to commit to one of the alternative paths. The TP is labelled and we compare the RP amplitude between the TP move vs other moves in the sequence.

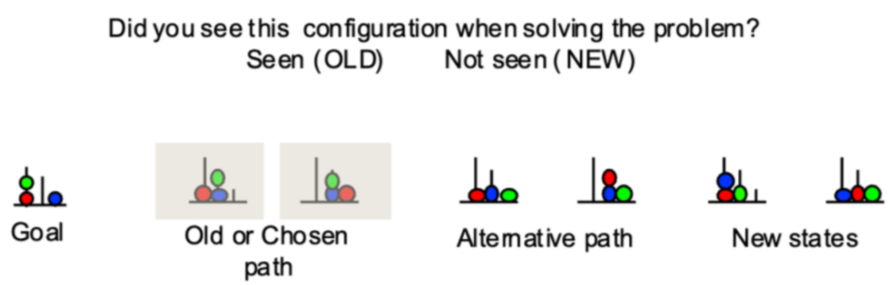

Figure 2: Experimental paradigm of the memory task: The participants were presented with 7 configurations (one at a time) and were asked whether they had seen this configuration before. They can either respond OLD if they think they had encountered the configuration before or NEW if they think they had never seen it.

3.3. Behavioral data analysis

All analyses were performed by means of the statistical software R (4.0.3) and the lme4 package (Bates et al., 2014).

To analyze the results of the memory test, we first derived the accuracy of the goal result. We assumed that if participants were unable to identify the goal correctly, they were not actively engaging in the task and their answers were conceivably arbitrary. Participants with a goal accuracy lower than 50% were automatically excluded from the following analysis. Then, we proceeded to perform a d prime analysis to identify those participants that were arbitrarily hitting random buttons all the way through without paying attention to the task (excluded participants=1). The signal detection theory (SDT), relying on d prime calculation allows us to distinguish those meaningful results from those random selections from inattentive participants. (Macmillan and Creelman, 1991). D'prime is a measure of an individual's ability to detect signals, it is the difference between the false alarm rate and the hit rate. The higher the d' prime value the better the performance, while value 0 means the subject is most likely just 'guessing'. SDT has been a common theoretical framework to analyze the behavioral response of humans. It takes the assumption that all stimuli are noisy and will result in overlapping trials of signal and noise distributions. Applying SDT to this study can help us evaluate the ability of participants to detect the alternative of solving the task. There are 4 possible outcomes depending on the answer given by the subject and the nature of the problem: 'hit' or when the subject correctly answered 'new' to 'new' or 'alternative' memory configuration; 'miss' when the subject answered incorrectly 'old' to a 'new' or 'alternative' configuration shown, 'false alarm' when the subject answers 'new' to a configuration that is 'old' or 'goal'; and correct rejection when the answer is 'old' in the actual 'old' configuration.

	"Old"	"Alternative"
"Old"	HIT	MISS
"Alternative"	FALSE ALARM	CORRECT REJECTION

	"Old"	"New"
"Old"	HIT	MISS
"New"	FALSE ALARM	CORRECT REJECTION

Figure 3: Signal Detection Theory (SDT) was applied to evaluate the subject's response. The 'Old' trials were split into half and used to compute with half of it the 'Alternative' d prime and with the other half 'New' d prime. Depending on the subject's answer to 'Old' or 'New' we classified the results into 4 categories: Hit, Miss, False alarm and Correct Rejection.

The uncertainty arising from external stimuli is called perceptual noise, when the noise increases, the accuracy will decrease, meaning that response states are not able to differentiate between noise and signal trials. The false alarms, that is responding incorrectly 'Seen' to the Unseen configurations, while conversely the hit rate would be that of the subject responding correctly 'Seen' to a Seen configuration. With the memory test, we want to examine whether in the memory test, the unseen alternative stimuli would raise a response bias, by having greater number of false alarm rates compared to the unseen new stimuli.

D prime relies on prior calculation of the z scores of both hit rate and false alarm rate, followed by the subtraction between the resulting values. In order to carry out SDT analysis we need to split "Old" trials in half and analyse half of it with "Alternative" and the other half of "Old" with "New". With this, we obtain a false alarm rate and hit rate for both Alternative vs old condition and new vs old condition. Finally, to calculate the d prime we compute the z score of hit rate and false alarm rate and subtract between the results. The d primes we obtain are conveying the likelihood of the participant to mistakenly answer "Old" to "Alternative" compared to that of "New". The estimate of the d prime can be anything between positive, zero and negative, with a higher d prime indicating a higher sensitivity to the stimuli. To ensure that there were no biases caused by the old trial distributions pairing, we carried out a total of 3 randomizations. Then, we went on to perform a paired 2 sample t-test between the d prime of "Old to alternative" with the d prime population mean of "Old to New".

EXPERIMENT 2

3.1. Participant recruitment for EEG

Participants were recruited using the UCL psychology subject pool. Requisites of sign up were: age between 18-40 right handed, normal or corrected to normal vision, having no history of psychiatric/psychological disorders or neurological disorders/injuries, including seizures, or any family history of seizures/epilepsy. Participants have to confirm that they are not taking any psychoactive medication, have no history of drug abuse and used any recreational drugs or alcohol 12 hours before the testing session.

Out of 22 participants (19 females) took part in the experiment. Participants are compensated for their time with an hourly rate of 9.5 pounds plus a bonus depending on the performance, which is reimbursement approved by the institution. Also, the experimental design and procedure followed the UCL ICN ethics committee (ICN-PH-PWB-20-02-2014c SAME AS BEFORE). Every participant was invited to a single EEG session. Prior to beginning the task, participants were asked to sign an informed consent form and were verbally instructed of the rules of the task.

3.4. EEG task and procedure

The EEG task was programmed in Matlab R2014b using Psychophysics Toolbox-3 extension. (Brainard, 1997; Kleiner et al., 2007; Pelli, 1997). And the "Tower of London" problems-solving tasks were designed with the help of tower tool extension. The ToL problems were modified from the original version (ToL, Shallice 19823) which was created with the initial purpose of examining the deficits of patients with frontal cortex damage. During the task, participants had to move the 3 colored pegs transforming the start configuration to the goal configuration in the minimum number of moves possible.

The ToL problems were selected to fit the different experimental conditions we want to examine. In stimuli problems, they were mainly self-generated moves where the participant had to find the solution on their own, and in stimuli moves, on the other hand, they followed guided instructions on how to solve the problem. Then, we selected ToL problems that included either 3 or 4 minimum moves to solve. And finally, TP is the bifurcation design where participants commit to one of the solving paths. To study the effect of TP on the RP amplitude, we selected both problems with and without TP of the same sequence length.

Participants completed a total of 6 blocks, which were randomized and presented in different order each time. Every block had 28 trials so a total of 168 trials were completed by each participant. During training part, participants completed a sample block to get familiar with the task and check they understood the instructions.

3.5 EEG Recording

Participants were placed in an electrically shielded chamber, and EEG was recorded from 26 scalp sites and amplified using g.Recorder (G.tec, medical engineering GmbH, Austria). While participants were playing with the behavioral task. The EEG cap consisted of 26 electrodes, including FZ, FCZ, CZ, CPZ, PZ, POZ, FC1, FC2, C1, C2, CP1, CP2, F3, F4, F7, F8, C3, C4, CP5, CP6, FC5, FC6, P3, P4, O1, O2. Also, we stuck HEOG bipolar electrodes to the outer canthi of each eye, VEOG above and below the right eye, and A1 and A2 electrodes to the mastoids behind the ear. Using active electrodes (g.LADYbird) fixed to an EEG cap (g.GAMMAcap) according to the extended international 10/20 system. EEG data were acquired using a g.GAMMAbox and g.USBamp with a sampling frequency of 256 Hz and 0.01 Hz high-pass and 100 Hz low-pass online filters. All electrodes were referenced to the right ear lobe. Vertical and horizontal electro-ocular activity was recorded from electrodes above and below the right eye and on the outer canthi of both eyes.

3.6 EEG Pre-processing

EEG data preprocessing was performed in Matlab R2021a (MathWorks, MA, USA), using the EEGLAB software package version 2021.0 (Delorme & Makeig, 2004).
All continuous EEG and EOG data were bandpass filtered with 0.01 Hz high pass filter. These filters were applied off-line using a 5th order Butterworth filter with zero phase shift. The data were down sampled with a rate of 200 Hz and filtered with a 30Hz low pass filter. Scalp and eye Electrodes were re-referenced to average of mastoid electrodes. The EEG recording were locked to the keypress, so each individual RP is believed to fall in epochs 2s before the keypress event and 1s after it..

Baseline correction was performed using a 10ms window centered on the time of the action [-5ms to +5ms relative to keypress] (Khalighinejad et al, 2018). Conventionally, RP recordings are baseline-corrected by subtracting the average signal value during an arbitrary-chosen window before the time of the action (for example, 2.5 until 2 s before action). However, this involves the questionable implicit assumption that RPs begin only in the 2 s before action onset (Verbaarschot et al., 2015). Our baseline centered on the time of the action avoids making any assumption about how or when the RP starts. This is particularly important in the present study considering the significant by-design difference in the preparation time preceding the action (target time ranging from 2.25ms to 7.25ms).

Independent Component Analysis (ICA) was computed on the continuous data using the EEGLAB runica algorithm (Delorme et al., 2007). Vertical and horizontal eye movement components were identified by visual inspection and removed from the signal.

Finally, artefact rejection was performed by removing all epochs with >120 µV fluctuations from baseline.

3.7 EEG Data Analysis

Analyses were run using the statistical software R (4.0.3) and the lme4 package (Bates et al., 2014). Also, we used Jamovi for data analysis and the creation of plots.

Results

3.4. Behavioral Test Results

The computation of the d prime in the memory test was used to evaluate whether the participant represented the unchosen but plausible alternative. If this was the case, the number of false alarm rate would be significantly greater for the alternative compared to the new configuration. To analyze this, We performed a 2 sample-t test. The d prime of the "New vs old" configurations results in an average 0.268 ±0.655 (mean ± SD), while the "Alternative vs Old" configurations is 0.432±0.780 (mean ± SD). A Shapiro Wilk Test was carried out to examine whether the population of data collected followed a normal distribution. The significance value of the Shapiro Wilk test (W=0.874, p-value=0.004) . A p-value<0.05 shows that we reject the null hypothesis, and the sample is data does not follow normal distribution. The visualization of the QQ-plot shows that the data is skewed to the right, which suggests that the data does not follow normal distribution. P-value<0.05 and rightly skewed QQ plot both indicate that we may need to use a non-parametric test, such as Wilcoxon rank. In Signal Detection Theory, the variables used to calculate the d prime such as hit rate and false alarm are not continuous variables, thereby cannot have a normal distribution. The Wilcoxon test produced (W=272, p-value=0.013), a p-value<0.05 means that we have the evidence to reject the null hypothesis and prove that the d primes of the "Old to Alternative" vs the d prime of "Old vs New" trials were significantly different from each other. (Park et al, 2020)

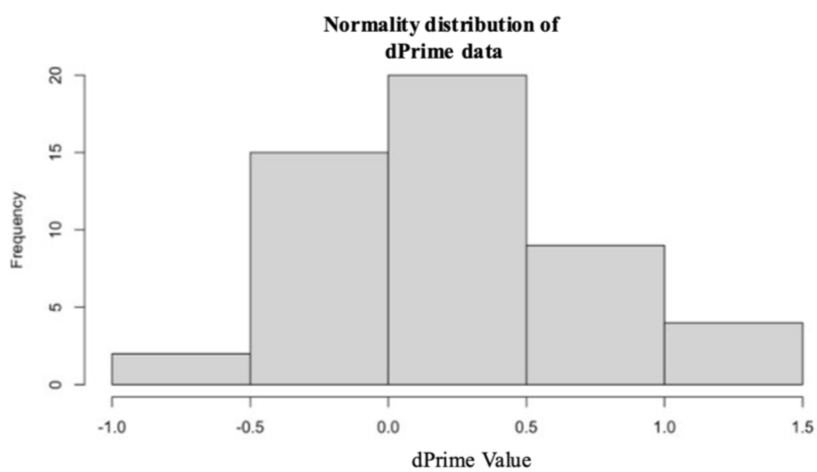

Figure 4: Normality distribution of d Prime data. The d prime Values for all the variables were normally distributed as visualized in this histogram.

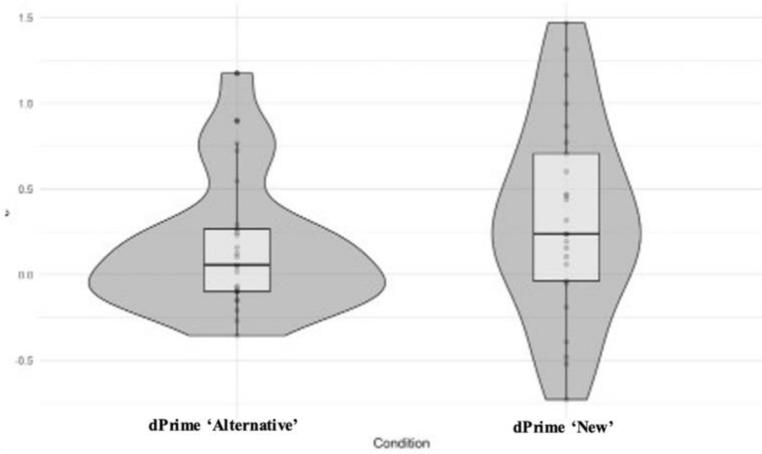

Figure 5: Box Plots of d Prime of Alternative vs d Prime of New conditions. The d Prime for 'Alternative' and 'New' show that participants are more biased towards misclassifying and responding incorrectly 'Old' to 'Alternative' trials compared to 'new' trials

4.2 EEG data

The EEG recording was divided into epochs [-1-0s] and averaged across all trials and then between all participants (N=20). We recorded RP amplitude 1000msec prior to movement onset of problems with and without the TP. The data was normally distributed so we did a paired 2 sample t-test (p-value 0.099, effect size=0.378). As p-values >0.05, the null hypothesis was not rejected in this occasion, indicating there was no significant difference in the RP amplitude of those trials with and without an alternative path. Then, we used a Shapiro-wilk test to measure the normality distribution (W=0.928, p-value=0.099). The significance level was greater than 0.05 so the data is normally distributed. For the comparison between stimuli with and without TP of the instructed problems, we obtained a

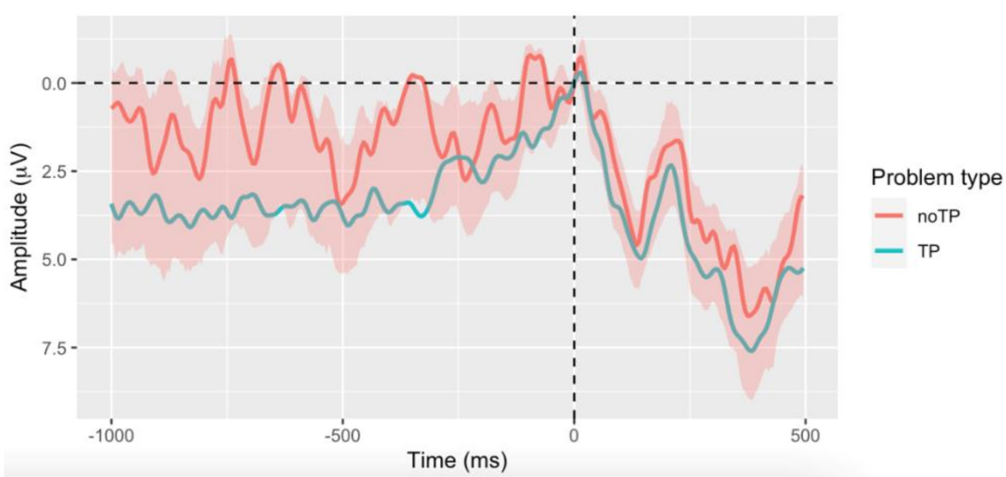

Figure 6: EEG epoch of Readiness Potential (RP) between problems that contain TP and noTP. Slight earlier onset of RP for TP problems compared to no TP, but RP amplitude difference not significant so the null hypothesis is not rejected

NEED TO SAY *SOMETHING HERE? ALSO, WHEN REPORTING RESULTS YOU NEED TO REPORT THE T VALUE AS WELL AS THE P VALUE* (p-value=0.729, effect size=0.0731). The Shapiro-Wilk test results in (W=0.974, p-value=0.793). Once again, for the problems with instructions, the comparison between TP and no TP was not significant, but the data was normally distributed.

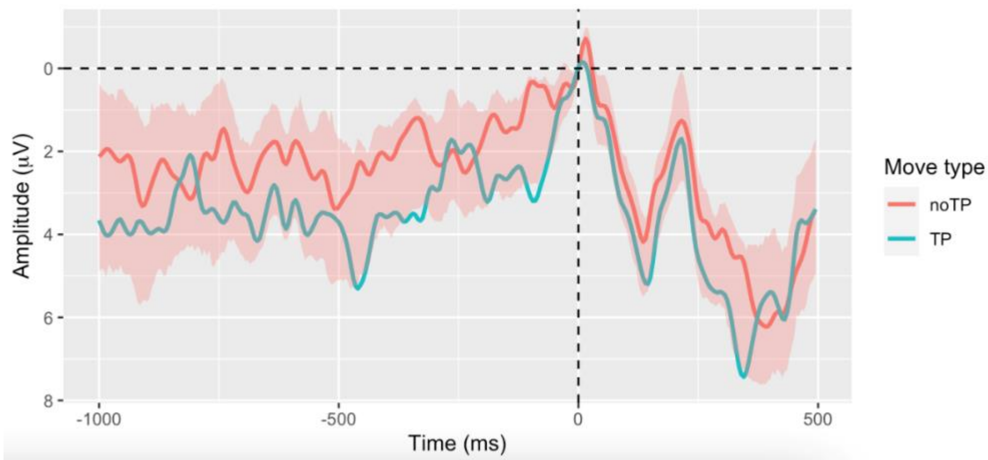

Figure 7: EEG epoch of Readiness Potential (RP) measured before first move of problems with and without alternative path. No significant difference in the RP amplitude between problems with (TP) or without alternative path (noTP)

On the other hand, we also compared the RP amplitude within one problem with TP, a 2-sample t-test was carried out comparing the actions preceding TP in the first step of the problem and any other moves after surpassing TP. In the Intention problems, we observed that the significance level was (Effect size=0.249, p-value=0.244). while for the control stimuli problems (Effect size=-0.0804, p-value=0.704). None of the results provide enough evidence to reject the null hypothesis, therefore results are not significant.

Be careful when you report the results: you need to report the test value as well as p value, say clearly whether it is significant or not and what it means (one at a time)

5. Discussion

5.1 'People care about the goals, but also about the means to achieve the goal'

A common discourse we have regarding voluntary action and problem solving is that of equifinality. The assumption is that the individual only cares about the 'goal' and not the 'means to achieve the goal'

(Haggard, 2019). This study aims to decipher whether we might have representation and access to the different choices in a decision, as well as whether the presence of choices makes a difference in the nature of motor actions involved in problem solving tasks. The major observation of the behavioral study shows that the memory representation of the ‘unchosen’ alternative choice is significantly greater than that of the New configurations. A disparity in the memory representation could be interpreted in 3 ways depending on the result we got when applying STD. Either it can be a result of a boosted memory representation, which results in a high number of false positives. This means that participants have a strong memory representation of the alternative choice as though they had seen it before, answering ‘Old’ although they had never encountered the configuration. We suggest that their brain perhaps represented both alternatives before making a decision. In the other scenario, if we get a lower number of false positives of the alternative compared to new, it could be explained using memory suppression. As we know, voluntary action involves a decision. It is suggested that this process of decision making, selecting a choice will cause inhibition of the ‘unchosen’ option. Previous fMRI studies have provided some evidence for this, believing that action selection and stopping are regulated by the pre-supplementary motor area (pre-SMA) and inferior frontal gyrus in order to prevent conflicts between the choices (Rae *et al*, 2014). If this theory was true, when a subject made a choice to take path A they would automatically inhibit and diminish the memory representation of path B to consolidate their choice. Finally, if there is no significance difference observed (p-value>0.05), it would mean that the subject just did not notice the alternative path. The subject is not even aware that they had the freedom of choice. If that was the case, how could we explain our actions as worthy of moral implications? “We are not aware that we have the freedom of choice the moment we decide”.

Instead, the results of the memory task show that the difference is significant ($p = 0.028$), while it also has evidence to argue against the memory suppression theory. We obtained a boosted memory representation with a high number of memory false positives, suggesting that the subject represents the alternative choice with a memory bias, believing that they had previously seen the ‘unchosen’ alternative configuration when they didn't. The participants were able to choose their movements within the experimental paradigm constraints.

Applying SDT to this study helps us evaluate whether participants have memory access to the process of solving the problem. If participants were representing both choices, they would incorrectly detect the alternative choice as ‘Seen’, because they would have stored both solving pathways within their memory. Otherwise, the participant could also not have represented the problem-solving process. In that case the alternative choice sensitivity should be similar to the new configurations, while also having a significant difference with the ‘Old stimuli’ The d’ prime or discriminability index computes how discriminable the signal is from the non-signal. And confirming that the difference between the d primes of alternatives and new trials are significant, using a paired 2 sample t-test, we can conclude that during the memory task, the false alarm rate for the alternative pathway is significantly greater than the new configuration. These results suggest that when solving a task, people do represent either conscious or unconsciously the task solving process, making them believe that had seen the ‘unchosen alternative’ when they had not. This interpretation does not align with the stochastic noise theory, which suggests that people are randomly drifted into one choice or another as a result of neural noise accumulation in the brain.

5.2 Turning Point does not make a difference to the amplitude of Readiness Potential

The RP amplitude, interpreted as a biomarker for voluntary action, is hypothesized to be dependent on the presence of a choice. Only when alternative choices are provided to the subject, they will be able to elicit a ‘self-initiated action’. A ToL task containing a Turning point is a conventional experimental paradigm resembling ‘decision making’ in daily activities. When faced with a bifurcation where both paths are equally as optimal, there is no reason for the participant to pick one alternative over the other, so the cognitive capacity to make a choice is thought to be correlated with a ‘self-initiated action’. We hypothesized that by showing that RP amplitude fluctuated between problems containing a TP and those with no TP. If the difference in the RP amplitude were to be significant it would provide evidence supporting RP as either a trigger or signal of that self-initiated action. However, EEG results of RP amplitude between TP vs no TP were not significant. So, we do not have enough argument against the view that the RP merely reflects accumulation of stochastic noise. The interpretation of RP as spontaneous fluctuations of background neuronal activity and all our decisions are swayed by this stochastic neural activity, is not rejected in this case (Schurger *et al*, 2012). Our other hypothesis was that the RP amplitude preceding the first TP move was going to be greater than any other move in the problem. As we expect that the action moves after surpassing TP becomes more a result of instinct rather than conscious action. Once again, the results show that the significancy difference between the 2 trials was not enough to reject the null hypothesis. We hypothesize that this may be due to the sample size, and if we increase the sample size in future studies, there might be a larger possibility of discovering a significance level developed between the 2 variables.

In this context, A potential future study would be using TP problems involving alternatives with different reward values. A previous study looked at how memory and decision making interact to shape the value of the alternative no-chosen option (Biderman & Shohamy, 2021). This study gives us insights on how people would respond to decisions which involve choices with different rewards. Would the magnitude of the RP reduce due to the increased presence of external stimuli in decision making? or else, could the RP amplitude increase caused by the aggregated number of factors the individual has to consider before making a choice? Answers to this question may offer a closer comparability to real life self-initiated actions people face in everyday life, especially those which involve conscious thought and consideration of pros and cons.

5. 3 Cognitive processing of alternative prior to onset of Readiness Potential

As the results showed no significant correlation between EEG recording of RP amplitude and presence of a TP. At first we may be struck by the incompatibility of the first and second experiments results, which do not seem to agree with each other. However, in this case, the hypothesis of our experiment we set could be fundamentally deceptive. It is likely that the RP has no correlation with the process of decision making. If that was the case, the representation of both choices before decision could have an even earlier onset than the RP move. That is ‘the person first represents both plausible choices and then commits to one of them when generating an action’. In this context, if the cognitive representation of the alternative is dissociated from the action generation process, it would make sense for the RP amplitude to be irresponsive to whether the problem has or does not have a TP.

The current neuroscience community believes that for each feeling, thought, action, there must be a chain of neural activity that is accompanying it, however what has made neuroscience so complicated to study is that this relationship between the mind and the brain has shown to be non-symmetrical, in other words there is not a direct relationship matching one type of neural activity with a behavioral outcome, a little like gene mapping. The result of a gene knockout may affect other areas of the molecular pathway which we did not intend to change. For this reason, it is a limitation of the study as we cannot know whether RP is solely a biomarker for Voluntary action or if it is also in control of other mental functions, however the timing evidence of RP clearly preceding each voluntary action, along with this experiment's results provide evidence that RP amplitude varies significantly depending on freely the action is generated.

In conclusion, applying problem solving activities to volition studies provides an experimental paradigm that shows better resemblance to those voluntary actions which we get involved in everyday life. Our results showed that the human represented both the end goal and the means by which to achieve the goal. As modern society operates based on the wide assumption that people are aware of their actions, the evidence that people are representing all the different alternatives that they could have taken is genuinely comforting. That is because people need to beware of their actions and 'freedom to choose' to be able to be blamable for their action consequences. This all comes back to the debate of the notion of free will and whether voluntary action is inherently based on the conscious intention of the subject or not.

Bibliography

Biderman, N., Shohamy, D. Memory and decision making interact to shape the value of unchosen options. Nat Commun 12, 4648 (2021). https://doi.org/10.1038/s41467-021-24907-x

Botvinick, M. M., Cohen, J. D., & Carter, C. S. (2004). Conflict monitoring and anterior cingulate cortex: an update. Trends in Cognitive Sciences, 8(12), 539–546. https://doi.org/10.1016/j.tics.2004.10.003

Rae, C. L., Hughes, L. E., Weaver, C., Anderson, M. C., & Rowe, J. B. (2014). Selection and stopping in voluntary action: A meta-analysis and combined fMRI study. *NeuroImage*, *86*, 381–391. https://doi.org/10.1016/j.neuroimage.2013.10.012

Fried, I., Haggard, P., He, B. J., & Schurger, A. (2017). Volition and Action in the Human Brain: Processes, Pathologies, and Reasons. The Journal of Neuroscience, 37(45), 10842–10847. https://doi.org/10.1523/JNEUROSCI.2584-17.2017

Frith, C. D., & Haggard, P. (2018). Volition and the Brain – Revisiting a Classic Experimental Study. *Trends in Neurosciences*, *41*(7), 405–407. https://doi.org/10.1016/j.tins.2018.04.009

Fourneret, P., & Jeannerod, M. (1998). Limited conscious monitoring of motor performance in normal subjects. Neuropsychologia, 36(11), 1133–1140. https://doi.org/10.1016/S0028-3932(98)00006-2

Haggard, P. (2019). The Neurocognitive Bases of Human Volition. Annual Review of Psychology, 70(1), 9–28. https://doi.org/10.1146/annurev-psych-010418-103348

Hari, S. D. (2018) "Voluntary Action, Conscious Will, and Readiness Potential", Cosmos and History: The Journal of Natural and Social Philosophy, 14(1), pp. 56–70. https://cosmosandhistory.org/index.php/journal/article/view/683

Kaller, C. P., Rahm, B., Spreer, J., Weiller, C., & Unterrainer, J. M. (2011). Dissociable Contributions of Left and Right Dorsolateral Prefrontal Cortex in Planning. Cerebral Cortex, 21(2), 307–317. https://doi.org/10.1093/cercor/bhq096

Khalighinejad, N., Schurger, A., Desantis, A., Zmigrod, L., & Haggard, P. (2018). Precursor processes of human self-initiated action. *NeuroImage*, *165*, 35–47. https://doi.org/10.1016/j.neuroimage.2017.09.057

Kornhuber HH, Deecke L (1990) Readiness for movement — The Bereitschaftspotential story. Citation Classic Commentary. Current Contents Life Sciences 33 (4): 14.

Libet, B., Gleason, C. A., Wright, E. W., & Pearl, D. K. (1983). Time of conscious intention to act in relation to onset of cerebral activity (readiness-potential). Brain, 106(3), 623–642. https://doi.org/10.1093/brain/106.3.623

Macmillan, N. A., & Creelman, C. D. (2004). *Detection Theory*. Psychology Press. https://doi.org/10.4324/9781410611147

Mineva, A., & Popivanov, D. (1996). Method for single-trial readiness potential identification, based on singular spectrum analysis. Journal of Neuroscience Methods, 68(1), 91–99. https://doi.org/10.1016/0165-0270(96)00012-X

Nachev, P., Rees, G., Parton, A., Kennard, C., & Husain, M. (2005). Volition and Conflict in Human Medial Frontal Cortex. Current Biology, 15(2), 122–128. https://doi.org/10.1016/j.cub.2005.01.006

Nguyen, V. T., Breakspear, M., & Cunnington, R. (2014). Reciprocal Interactions of the SMA and Cingulate Cortex Sustain Pre Movement Activity for Voluntary Actions. The Journal of Neuroscience, 34(49), 16397–16407. https://doi.org/10.1523/JNEUROSCI.2571-14.2014

Schurger, A. (2018). Specific Relationship between the Shape of the Readiness Potential, Subjective Decision Time, and Waiting Time Predicted by an Accumulator Model with Temporally Autocorrelated Input Noise. Eneuro, 5(1), ENEURO.0302-17.2018. https://doi.org/10.1523/ENEURO.0302-17.2018

Sumner, P., Nachev, P., Morris, P., Peters, A. M., Jackson, S. R., Kennard, C., & Husain, M. (2007). Human Medial Frontal Cortex Mediates Unconscious Inhibition of Voluntary Action. Neuron, 54(5), 697–711. https://doi.org/10.1016/j.neuron.2007.05.016

Sumner, P., & Husain, M. (2008). At the Edge of Consciousness: Automatic Motor Activation and Voluntary Control. *The Neuroscientist*, *14*(5), 474–486. https://doi.org/10.1177/1073858408314435

Szczepanski, S. M., & Knight, R. T. (2014). Insights into Human Behavior from Lesions to the Prefrontal Cortex. *Neuron*, *83*(5), 1002–1018. https://doi.org/10.1016/j.neuron.2014.08.011

Park, HD., Barnoud, C., Trang, H. et al. Breathing is coupled with voluntary action and the cortical readiness potential. Nat Commun 11, 289 (2020). https://doi.org/10.1038/s41467-019-13967-9

Travers, E., Friedemann, M., & Haggard, P. (2021). The Readiness Potential reflects planning-based expectation, not uncertainty, in the timing of action. *Cognitive Neuroscience*, *12*(1), 14–27. https://doi.org/10.1080/17588928.2020.1824176

Vierkant, T., Deutschländer, R., Sinnott-Armstrong, W., & Haynes, J.-D. (2019). Responsibility Without Freedom? Folk Judgements About Deliberate Actions. *Frontiers in Psychology*, *10*. https://doi.org/10.3389/fpsyg.2019.01133

Zhang, L., Zhang, R., Yao, D., Shi, L., Gao, J., & Hu, Y. (2020). Differences in Intersubject Early Readiness Potentials Between Voluntary and Instructed Actions. Frontiers in Psychology, 11. https://doi.org/10.3389/fpsyg.2020.529821